

#ASLI ENGINEERING

What is System Design

SWIPE

BY

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What is System Design?

Set of requirements

- ↳ decide architecture
- ↳ decide components
- ↳ decide modules

} how they interact with each other

↓

To solve the problem*

"Product Development" ← almost

Why is it so popular?

Every single "tech" product is a "system" this has been "designed"

companies are building product and need you all to design it

Why understanding system design is important?

- this is what people do at work
- everything is practical
- you can see it from Day 1 of joining the company
- once you grow in your career,

  you will spend 80% of time doing this.

System design is hence relevant for literally everyone!!

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Side effects of System Design

- it will make everything uninteresting (else)
- you solve real problems, not something made up
- break down problem statement
- re-wires your brain to think in a structured way
  - ↳ considering all possible cases to deliver a great user experience!

What will we do when we design a system?

- break down problem statement into solvable sub-problems
- decide on key components and responsibilities
- decide on boundaries of each component
- touch upon key challenges in scaling it
- make our architecture faulttolerant and available

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